

Ministère de l'Enseignement Supérieur et la Recherche Scientifique

Faculté des Sciences Exactes et Informatique

Département d'Informatique

Tp Module:

NoSQL & Big Data



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Exemple Simple



NoSQL

01

Initiation au NoSQL

Définition

Nosql: **N**ot **O**nly **S**QL.

- Système de bases de données non relationnelles.
- Possibilité de mise à l'échelle.
- Tolérance au partitionnement.
- Haute disponibilité.
- Rapidité d'exécution des requêtes.



Quelle est la difference entre SQL et NoSQL?

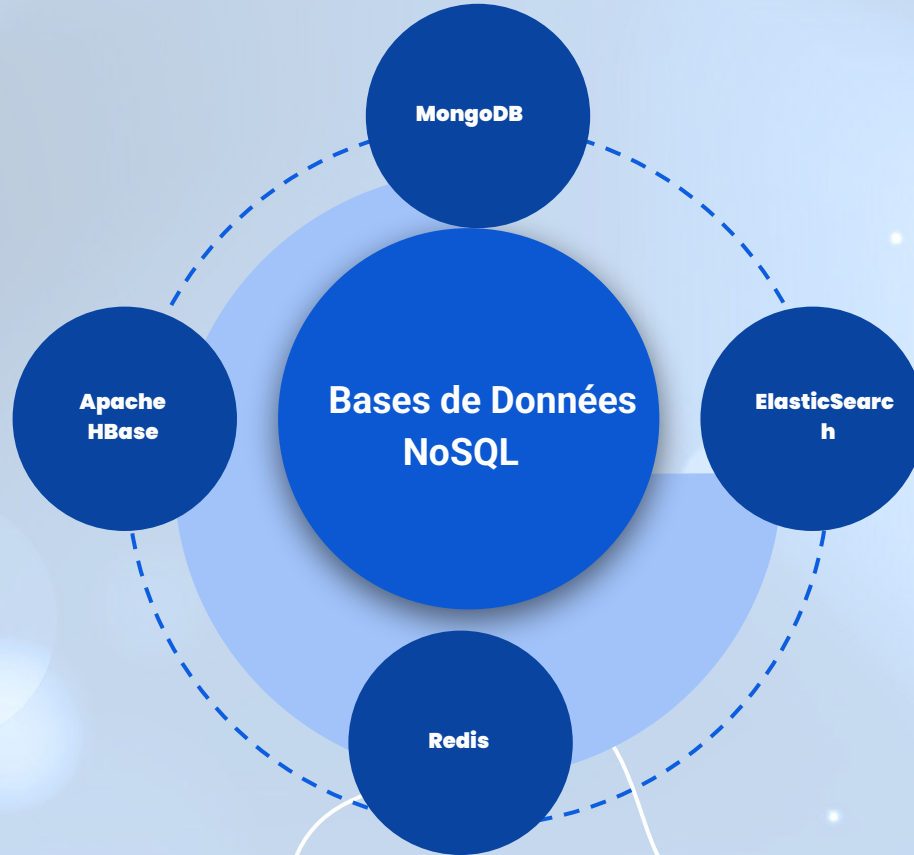
	SQL	NoSQL
Type	Relationnelle	No Relationnelle
Données	Structurées	Non Structurées
Schéma	Statique	Dynamique
Scalabilité	Verticale	Horizontale
Langage	Structured Query Language	Un-Structured Query Language
Support	Support éditeurs	Support Communauté open-source



02

Bases de données NoSQL

Parmi les Bases de Données NoSQL:

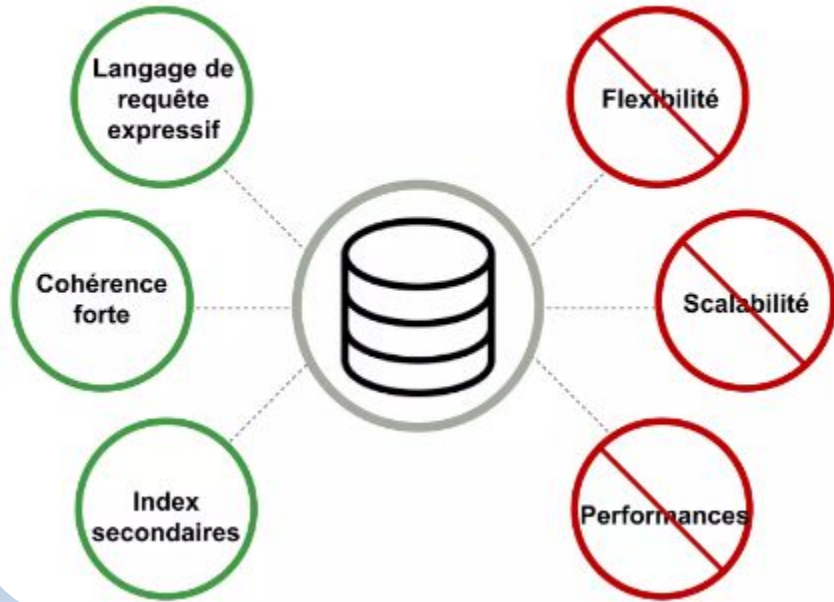




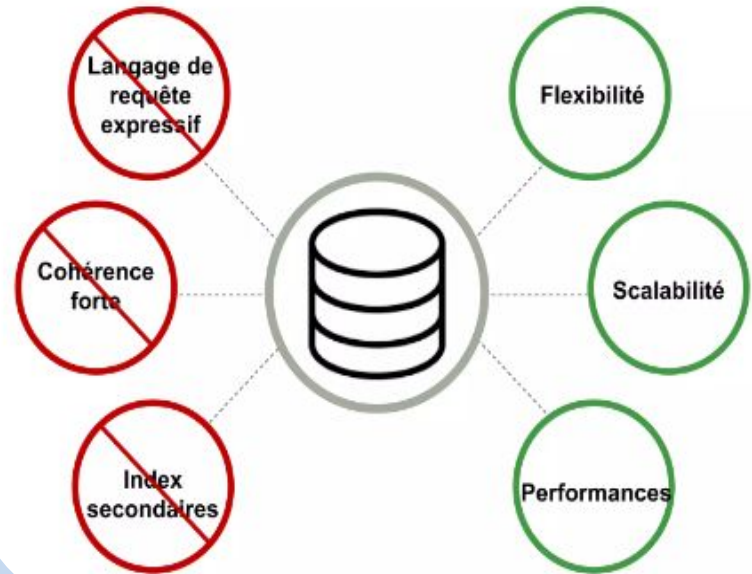
03

MongoDB

Base de données relationnelle



NoSQL



Base de données relationnelle

Langage de
requête
expressif

Cohérence
forte

Index
secondaires



Flexibilité

Scalabilité

Performances

NoSQL

04

Lien de téléchargement

<https://www.mongodb.com/docs/manual/installation/>

Guide d'installation

https://onlinehelp.coveo.com/fr/ces/7.0/administrator/installing_mongodb_on_a_server.htm#:~:text=D%C3%A9marrez%20l'interpr%C3%A9teur%20de%20commandes,%5Cbin%5Cmongo.exe.



05

Exemple 01

```
halla_guetarni@halla-guetarni: ~  
(base) halla_guetarni@halla-guetarni:~$ mongosh  
Current Mongosh Log ID: 653d0278fb40031fdf729d20  
Connecting to:      mongodb://127.0.0.1:27017/?directConnection=true&serverS  
electionTimeoutMS=2000  
Using MongoDB:      4.4.25  
Using Mongosh Beta: 0.12.1  
  
For mongosh info see: https://docs.mongodb.com/mongodb-shell/  
  
-----  
The server generated these startup warnings when booting:  
2023-10-28T13:12:16.392+01:00: Using the XFS filesystem is strongly recommend  
ed with the WiredTiger storage engine. See http://dochub.mongodb.org/core/prodno  
tes-filesystem  
2023-10-28T13:12:17.263+01:00: Access control is not enabled for the database  
. Read and write access to data and configuration is unrestricted  
-----  
> █
```



```
halla_guetarni@halla-guetarni: ~  
(base) halla_guetarni@halla-guetarni:~$ mongosh  
Current Mongosh Log ID: 653d0278fb40031fdf729d20  
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tes-filesystem  
2023-10-28T13:12:17.263+01:00: Access control is not enabled for the database  
. Read and write access to data and configuration is unrestricted  
-----  
  
> show dbs  
admin      147 kB  
config     111 kB  
local      73.7 kB  
> |
```

```
halla_guetarni@halla-guetarni: ~  
Using Mongosh Beta: 0.12.1  
For mongosh info see: https://docs.mongodb.com/mongodb-shell/  
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The server generated these startup warnings when booting:  
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tes-filesystem  
2023-10-28T13:12:17.263+01:00: Access control is not enabled for the database  
. Read and write access to data and configuration is unrestricted  
-----  
> show dbs  
admin      147 kB  
config     111 kB  
local      73.7 kB  
> use Etudiant  
switched to db Etudiant  
> show dbs  
admin      147 kB  
config     111 kB  
local      73.7 kB  
>   
>
```

Création d'une BDD


```
halla_guetarni@halla-guetarni: ~  
admin    147 kB  
config   111 kB  
local    73.7 kB  
> use Etudiant  
switched to db Etudiant  
> show dbs  
admin    147 kB  
config   111 kB  
local    73.7 kB  
> use Etudiant  
already on db Etudiant  
> db.Etudiant.insert(  
... {  
..... nom: "Mohammed",  
..... prenom: "Ahmed"  
..... }  
... )  
DeprecationWarning: Collection.insert() is deprecated. Use insertOne, insertMany  
, or bulkWrite.  
{  
  acknowledged: true,  
  insertedIds: { '0': ObjectId("653d03d54112da7f10915fb1") }  
}  
>
```

```
halla_guetarni@halla-guetarni: ~  
> show dbs  
admin      147 kB  
config     111 kB  
local      73.7 kB  
> use Etudiant  
already on db Etudiant  
> db.Etudiant.insert(  
... {  
..... nom: "Mohammed",  
..... prenom: "Ahmed"  
..... }  
... )  
DeprecationWarning: Collection.insert() is deprecated. Use insertOne, insertMany  
, or bulkWrite.  
{  
  acknowledged: true,  
  insertedIds: { '0': ObjectId("653d03d54112da7f10915fb1") }  
}  
> show dbs  
Etudiant   8.19 kB  
admin      147 kB  
config     111 kB  
local      73.7 kB  
> |
```

```
halla_guetarni@halla-guetarni: ~  
> db.Etudiant.insert(  
... {  
..... nom: "Mohammed",  
..... prenom: "Ahmed"  
..... }  
... )  
DeprecationWarning: Collection.insert() is deprecated. Use insertOne, insertMany  
, or bulkWrite.  
{  
  acknowledged: true,  
  insertedIds: { '0': ObjectId("653d03d54112da7f10915fb1") }  
}  
> show dbs  
Etudiant 8.19 kB  
admin     147 kB  
config    111 kB  
local     73.7 kB  
> db.dropDatabase()  
{ ok: 1, dropped: 'Etudiant' }  
> show dbs  
admin     147 kB  
config    111 kB  
local     73.7 kB  
> |
```